1. Application Features

1.1 User Signup

- Users can create an account by providing basic information such as name, email, and password, see Figure 2 for User Signup page.

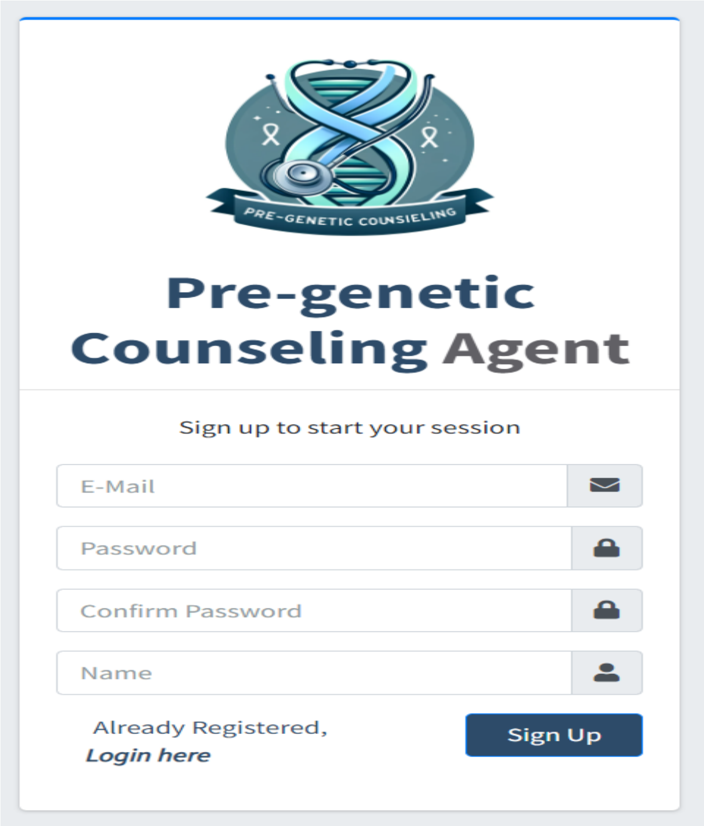

Figure 2: User Signup page

1.2 Introductory scripts/Videos

- Users are presented with a series of introductory script/videos explaining the application's purpose and how to use it. Introductory pages include the following:

1.2.1 Starting with welcome page. See Figure 3

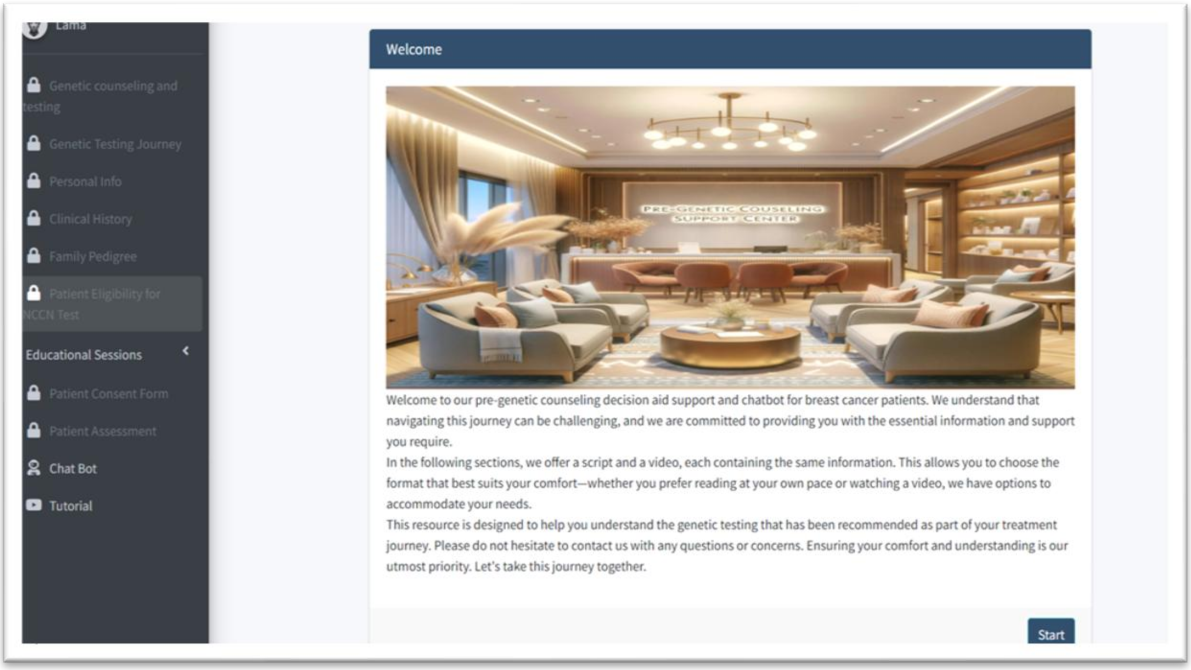

Figure 3: Welcome page

1.2.2 Introduction for Genetic counselling and testing page, presented as script and video. See Figure 4.

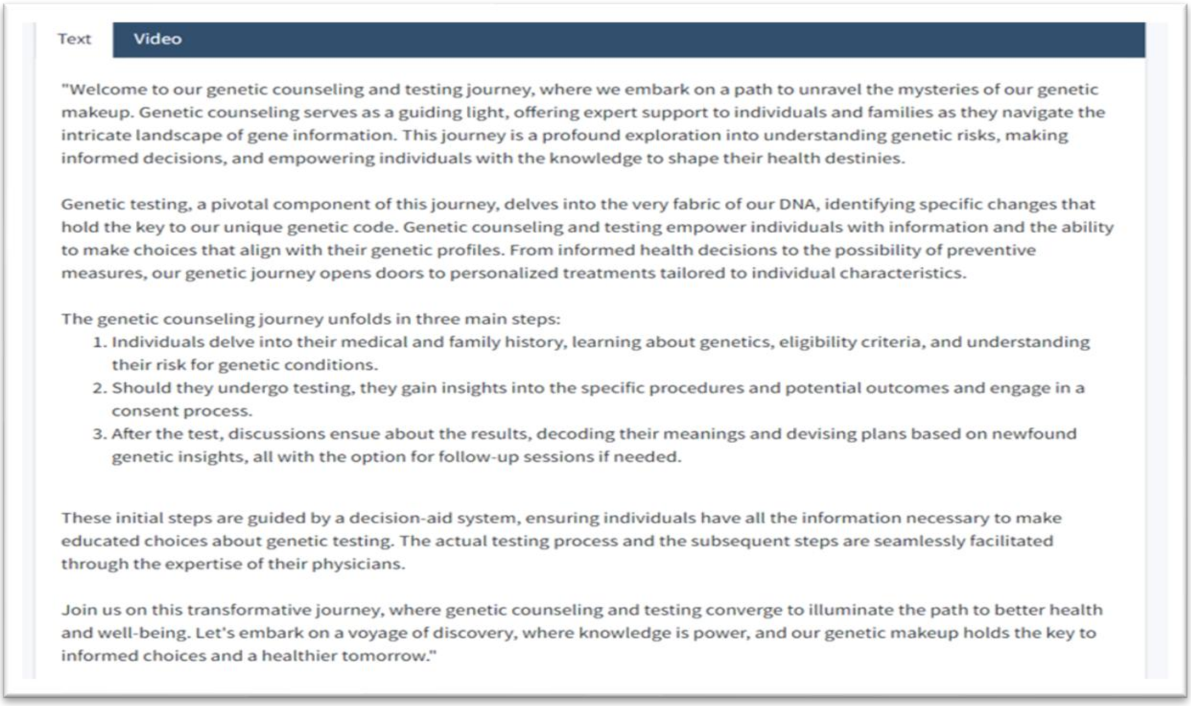

Figure 4: Introduction for Genetic counselling and testing page.

1.3 Summary for the DAS system framework, presented as script and video. See Figure 5.

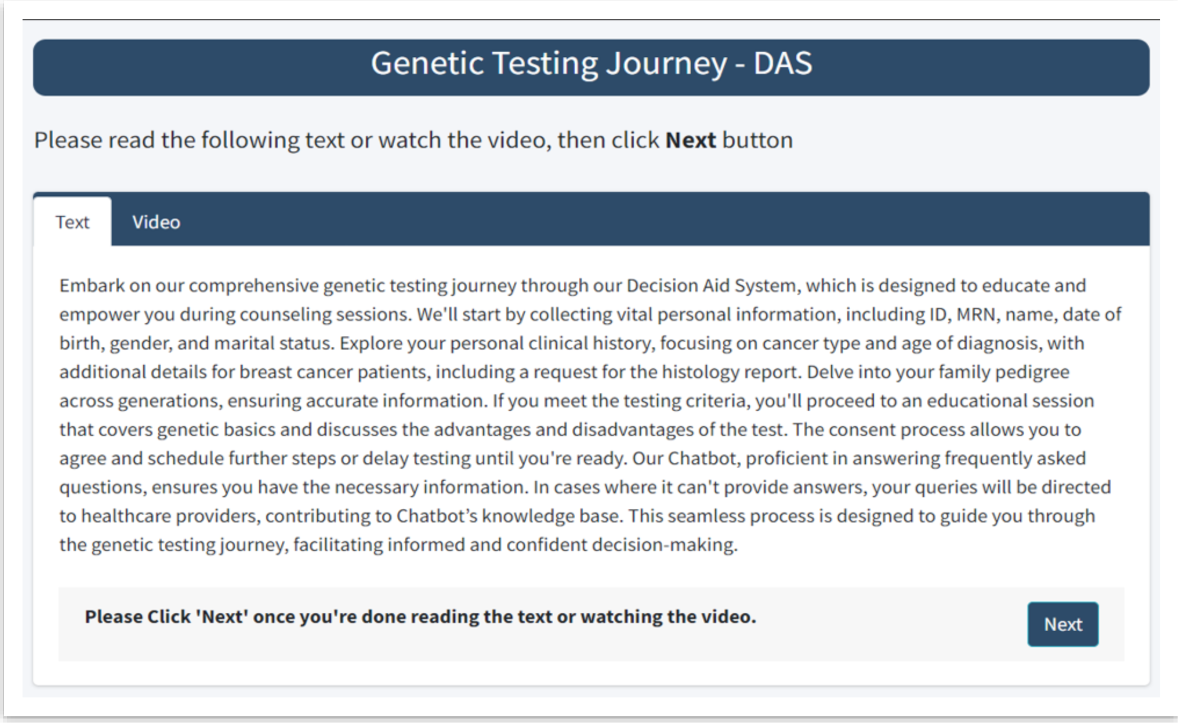

Figure 5: Summary for the DAS system framework-script

1.4 Personal Data Entry

Users input personal information, including age, gender, and medical history, See Figure 6 for personal data entry page.

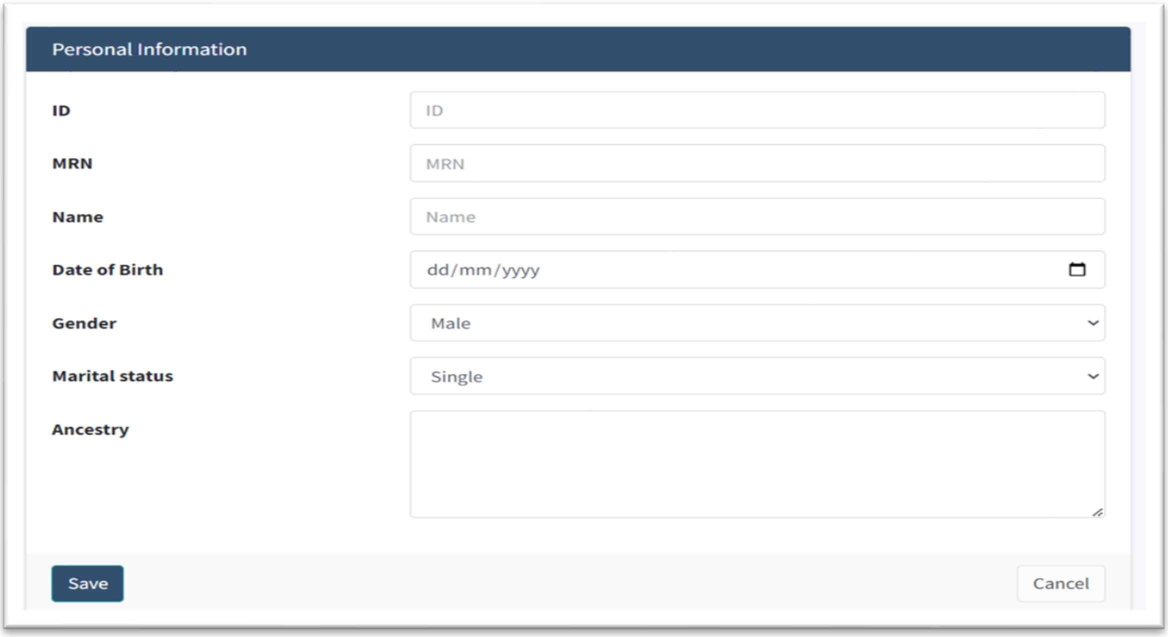

Figure 6: personal data entry page

1.5 Clinical History Input

Users provide detailed clinical history, including a histology report which contains information about tissue samples. See Figure 7 for Clinical History Input page

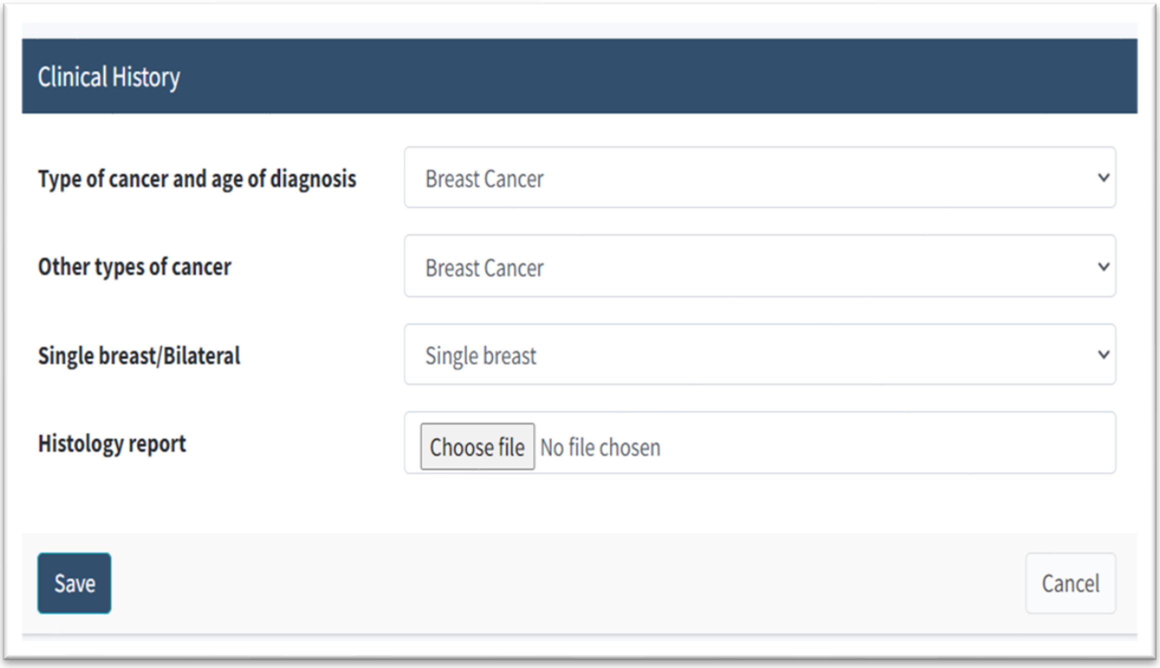

Figure 7: Clinical History Input page

1.6 Family Pedigree

Users may record family history by either drawing a pedigree in the platform or uploading an existing file. The pedigree is then analysed with AI tools that apply NCCN criteria to generate a genetic-risk assessment (see Figures 8a and 8b, Family Pedigree interface).

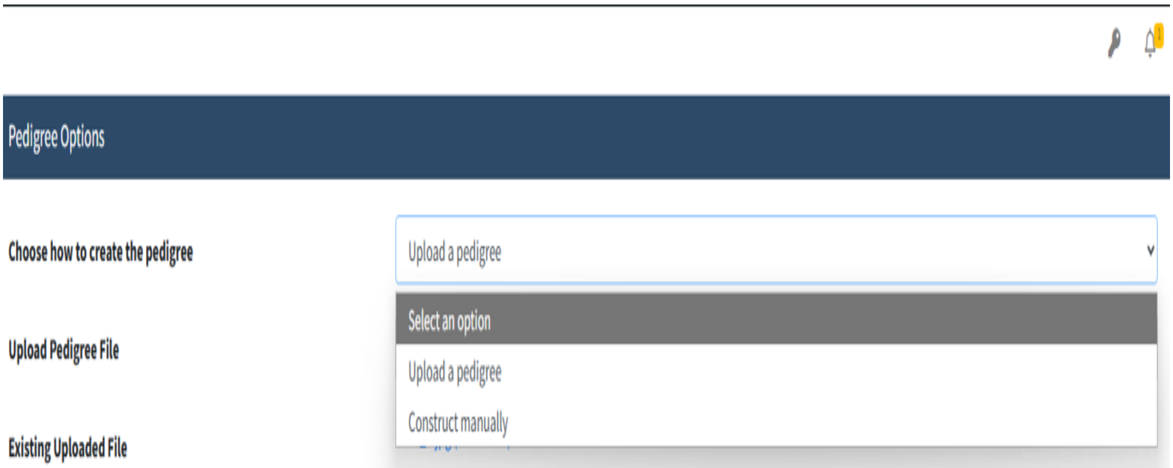

Figure 8a: Family Pedigree page

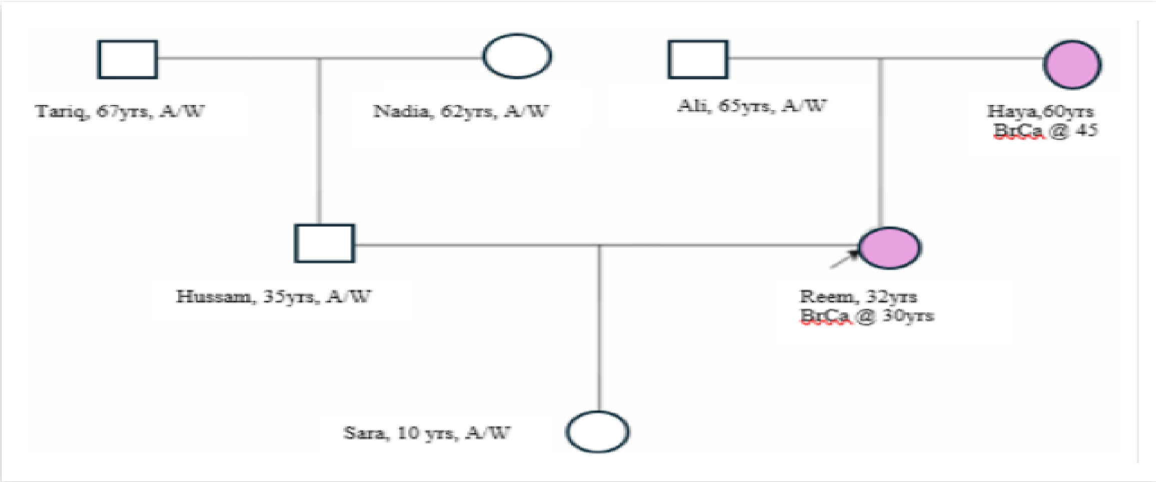

Figure 8b: Family Pedigree/construct manually

1.7 AI-Based Evaluation

The AI analyses the histology report and family pedigree to evaluate patient's eligibility for NCCN testing criteria. Results are displayed to the user with detailed explanations. See figure 9 for Patient Eligibility page.

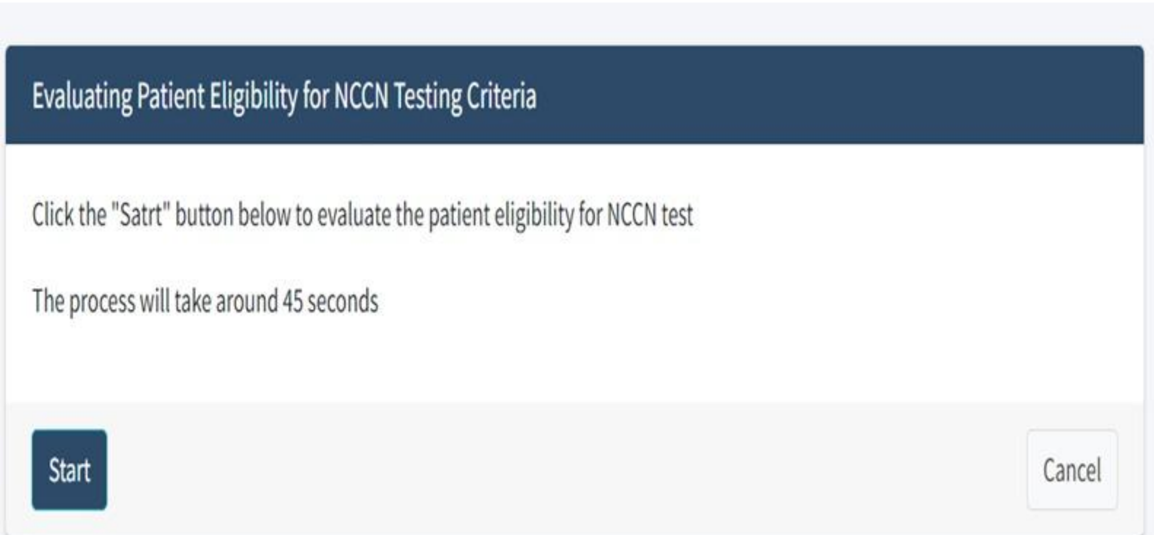

Figure 9: Patient Eligibility page

1.8 Educational session

Patients may attend the educational session only after they are confirmed eligible for genetic testing. User will introduce to the educational session scripts /videos that covers:

1.8.1 Concept of genetics and mode of inheritance, See Figure 10 for Concept of genetics and mode of inheritance.

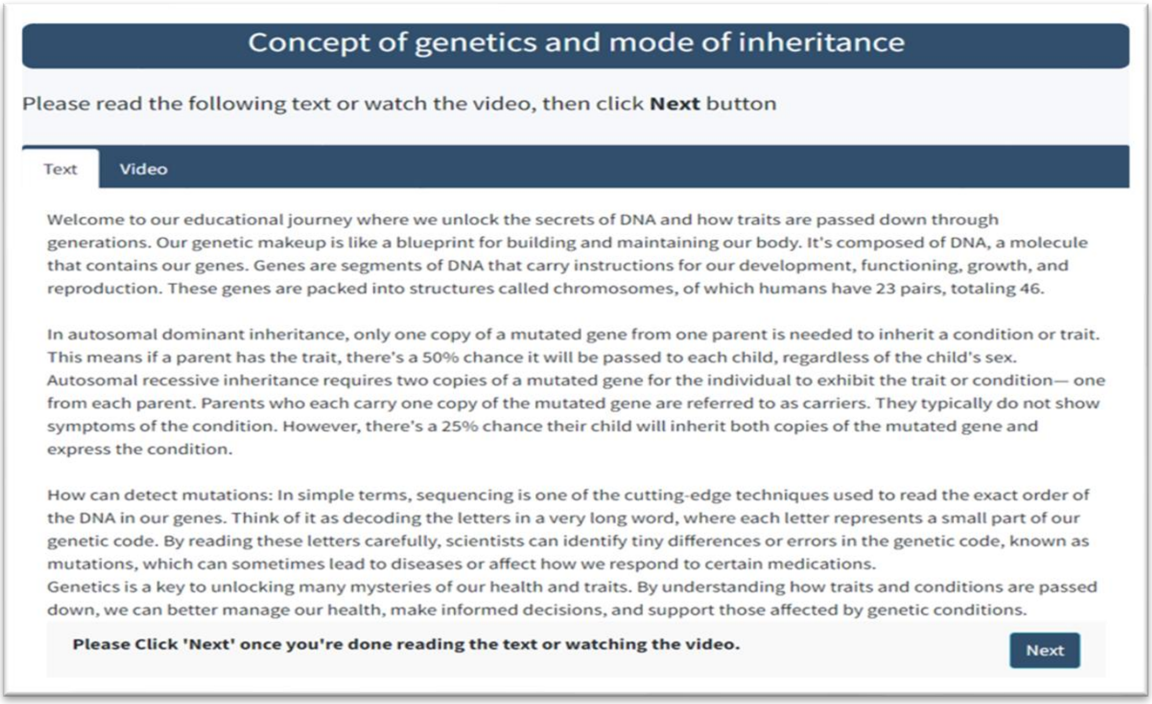

Figure 10: Concept of genetics and mode of inheritance.

1.8.2 Multigene panel and Outcome results. See figure 11 for multigene panel and outcome results.

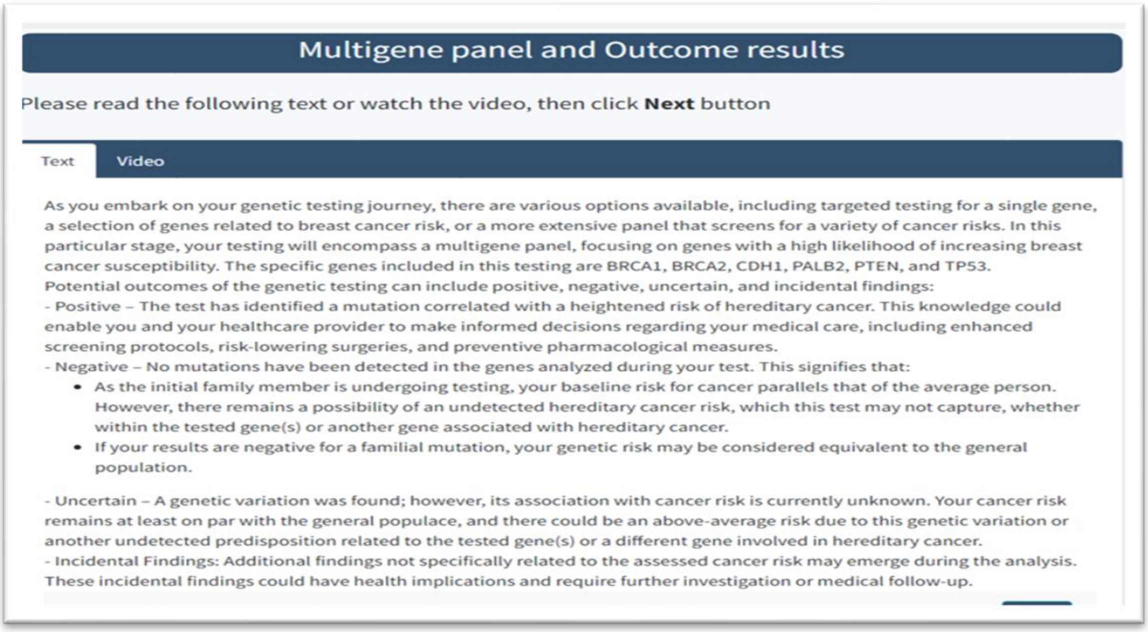

Figure 11: Multigene panel and Outcome results.

1.8.3 Benefits, Risks, and Limitations of genetic testing. See Figure 12 for Benefits, Risks, and Limitations of the genetic testing.

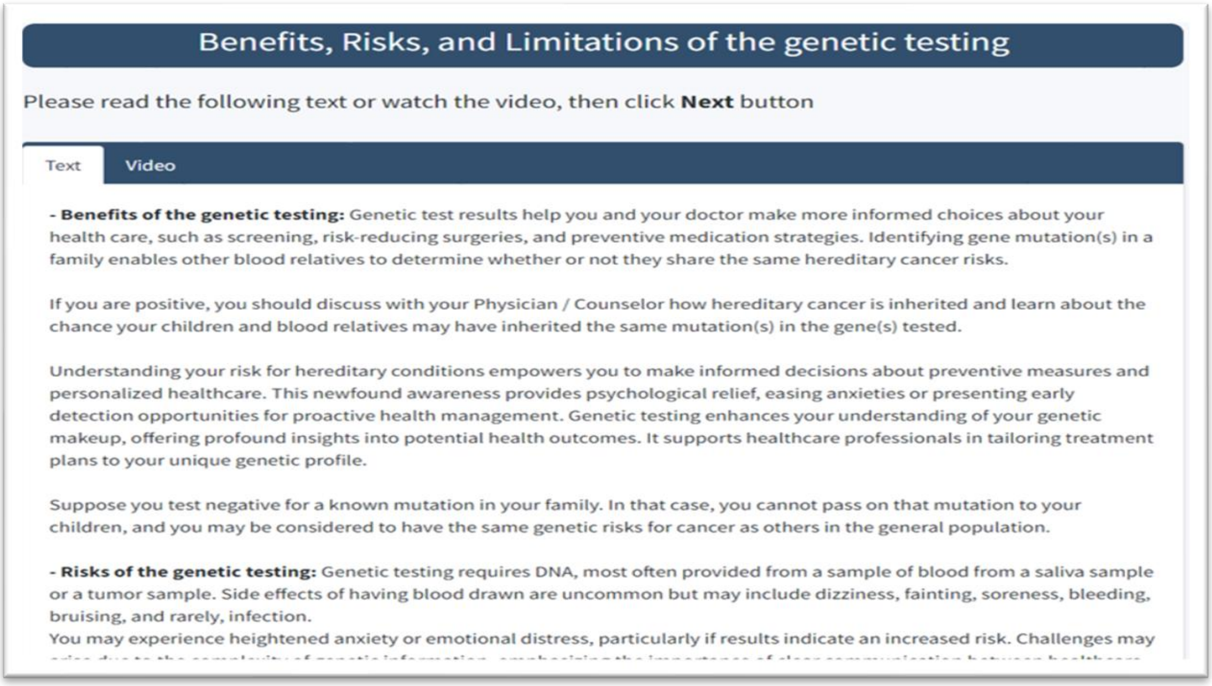

Figure 12: Benefits, Risks, and Limitations of the genetic testing.

1.9 Consent form and patient's assessments:

Once a patient chooses to undergo genetic testing, they can electronically sign the consent form and complete a 10–15-item questionnaire to assess their understanding of the procedure.

1.10 ChatBot

This OpenAI-powered chatbot, enriched with NCCN guideline content and other Frequently Asked Questions, can answer questions across the domains illustrated in figure 13. Patients navigate the application and interact via the Chatbot page, see figure 14.

Chatbot for pre-genetic counseling session:

I can help with the following type of information:

1. FAQs for High-Risk Breast Cancer Patients in pre-genetic counseling
    a. Personal and Family History pre-genetic counseling.
    b. Psychological pre-genetic counselling.
    c. Educational pre-genetic counselling.
    d. Clinical pre-genetic counselling.
    e. Sociological pre-genetic counselling.
    Physiological pre-genetic counselling.
    f. FAQs on high-penetrance breast cancer susceptibility genes: BRCA1, BRCA2, CDH1, PALB2, PTEN, TP53
2. Gene summary: Risk and Management
3. Genetic /Familial high-risk assessment and counseling
4. Multi-Gene testing
5. High penetrance Breast cancer/Ovarian Cancer Susceptibility Genes
6. BRCA related Breast/Ovarian cancer syndrome
7. Cancer risk management and screening
8. Pathogenetic/ likely pathogenic variants associated with Breast/Ovarian cancer
9. Genetic testing criteria

Figure 13: Information Domains Covered by the Pre-Genetic Counselling Chatbot

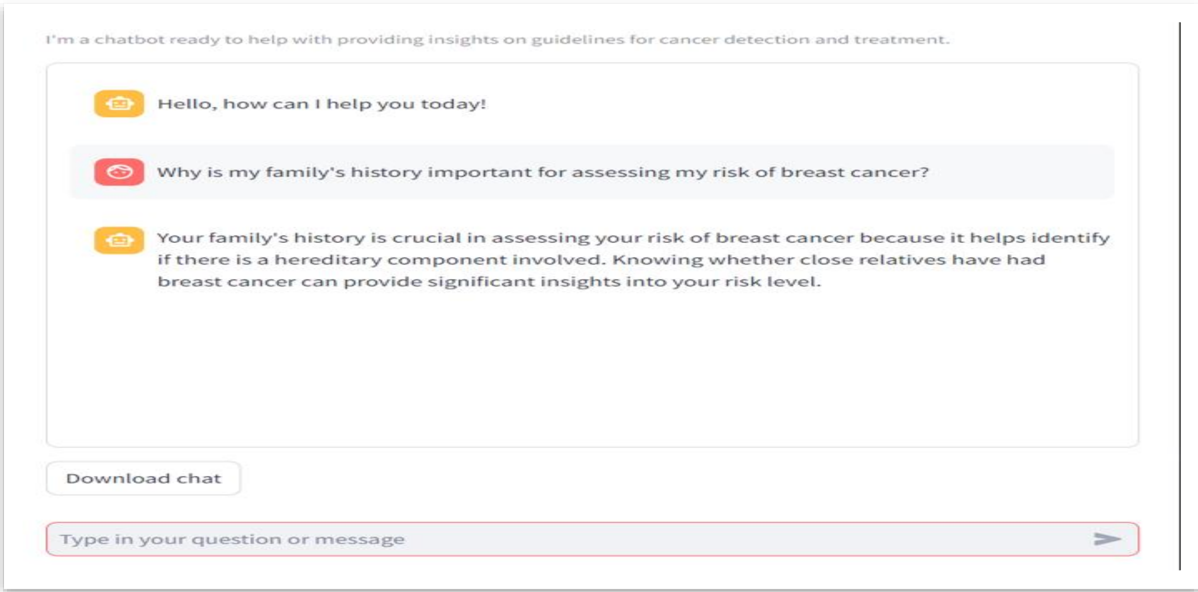

Figure 14: ChatBot page